The Optimization Landscape

When local information tells the global truth

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Two optimization problems

Linear and nonlinear model fitting

Both of these minimize a loss — L14 taught us that's what "fitting" is:

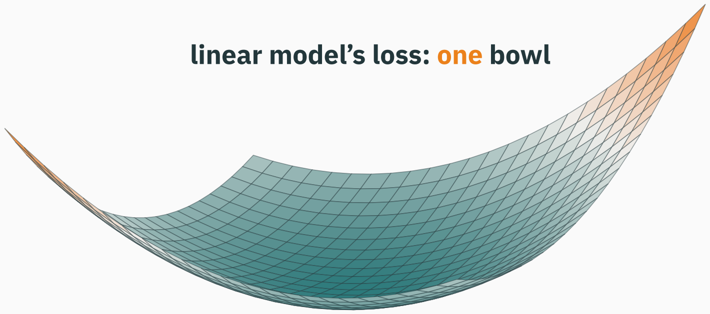
	fit a linear model	**train a deep network**
the code	`np.linalg.lstsq(X, y)`	loop of `loss.backward(); opt.step()`
knobs to tune	none	learning rate, schedule, init, batch size, ...
rerun on the same data	same answer, every time	depends on the seed
failure modes	—	divergence, plateaus, `nan` at step 617 (L2!)

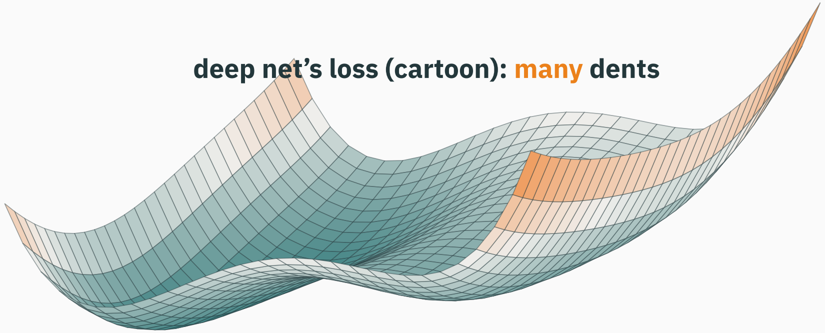
Both tasks **minimize a function of the parameters**. Which differences come from the algorithm, and which from the function's geometry?

Convex and non-convex objectives

V VISUAL

The difference is the **shape of the surface** each one walks on:





- Left: the objective has one basin; every local minimum is global
- Right: the objective can have several valleys, ridges, and saddle points

For a convex objective, every local minimum is a global minimum.

From estimation to optimization

L14 expressed parameter fitting as minimization:

$$\hat{\theta} = \arg\min_{\theta} \underbrace{\mathrm{NLL}(\theta)}_{\text{MSE, BCE, CE, } \ldots}$$

L14: “Fitting = minimizing NLL. Module 4 is the art of walking downhil

Module 4 studies this minimization problem in six stages:

L16 objective geometry → **L17** gradient descent → **L18** Newton methods → **L19–20** constraints and KKT → **L21** recognize LP/QP

This lecture develops the geometry used by the remaining optimization lectures.

Learning outcomes

By the end of this lecture you will be able to:

1. Use the optimizer's vocabulary: **objective, minimizer, stationary point, local vs global**.
2. Classify flat spots by the Hessian's **eigenvalue signs** — L9′s gallery, now on the job.
3. Test a **set** for convexity with segments, and a **function** with chords.
4. Certify convexity fast: $f'' \geq 0$ / $H \succeq 0$, plus the **sum, max, affine-composition** rules.
5. Prove ⭐ that every local minimum of a convex function is global.
6. Prove **least squares is convex** via $H = 2X^\top X \succeq 0$ (⭐) — and say what deep nets give up.

Local information available to an optimizer

The problem, stated once for the whole module

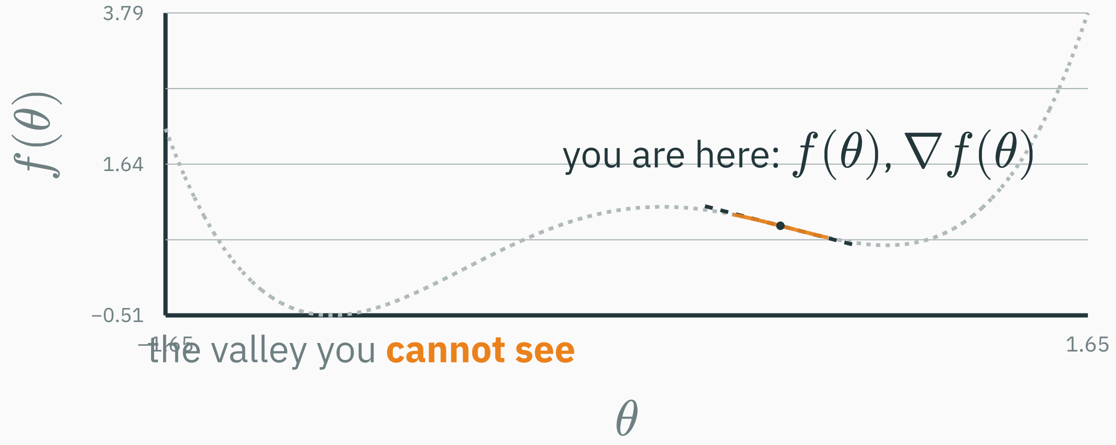
$$\min_{\theta \in \mathbb{R}^d} f(\theta)$$

- f: the **objective** — for us, almost always a loss / NLL (L14)
- a **minimizer** $\theta^\star$: an input where the smallest value is reached — $\theta^\star = \arg\min_\theta f(\theta)$; the **minimum** $f(\theta^\star)$ is the value there
- maximizing is the same problem upside-down: $\max f = -\min(-f)$ — L14 already negated log-likelihoods for exactly this reason, so **everything from now on is a minimization**

The optimization problem is $\min_\theta f(\theta)$; the following lectures differ in how they use information about f.

Function values, gradients, and Hessians

V VISUAL



At a current parameter value θ, an optimizer can evaluate three kinds of local information:

probe	what it is in ML	cost
$f(\theta)$	run the model, read the loss	cheap
$\nabla f(\theta)$	backprop (L8–L11)	cheap — one backward pass
$H(\theta)$	curvature matrix (L9)	d^2 entries — rarely affordable

Why not just look everywhere?

Because θ lives in $\mathbb{R}^d$, and d is not 2. Grid-search with a modest 10 values per coordinate:

model	d	grid evaluations 10^d
line $wx + b$	2	$10^2 = 100$ — trivial
tiny MLP	10^3	10^{1000} — no.
GPT-class model	10^{11}	$10^{10^{11}}$ — the observable universe has $\sim 10^{80}$ atoms

Exhaustive search is impossible at ML scale; practical methods use local evaluations of f, ∇f, and sometimes H.

Local derivatives describe a neighbourhood. Convexity is the additional global property that makes those local statements sufficient.

Taylor models use local information

L9 built "the formula every optimizer lives on" — today it goes to work:

$$f(\theta+\delta) \approx \underbrace{f(\theta)}_{\text{probe 1}} + \underbrace{\nabla f(\theta)^\top \delta}_{\text{probe 2: the local plane}} + \underbrace{\frac{1}{2}\delta^\top H(\theta)\,\delta}_{\text{probe 3: the local bowl or saddle}}$$

- L17 trusts the first two terms for one small step — **gradient descent**
- L18 trusts all three and jumps — **Newton's method**
- Today's question is prior to both: when do **local** conclusions ("it's flat here", "it curves up here") say anything about the **global** landscape?

Stationary points: where the search can stop

Stationary points

While $\nabla f(\theta) \neq \mathbf{0}$, the Taylor plane is tilted; a sufficiently small step against the gradient lowers f (L17 makes this precise).

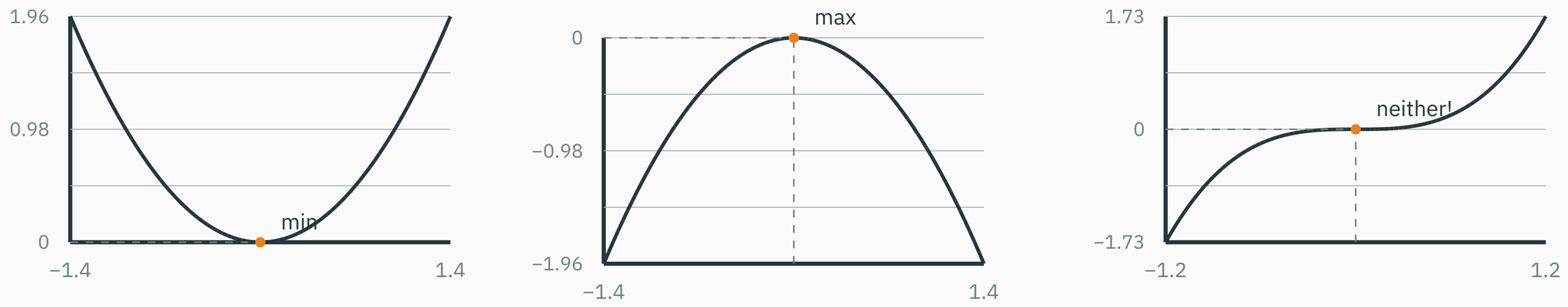
$$\text{the search can stop only where} \quad \nabla f(\theta) = \mathbf{0} \quad \text{— a } \textbf{stationary point}$$

- Also called a **critical point**; the first-order model is flat there
- But **flat is not the same as bottom** — L7 met smiles, frowns and x^3's fake-out in 1-D

> The gradient test finds candidate optima. The Hessian distinguishes minima, maxima, and saddle points.

Classifying stationary points in one dimension

V VISUAL



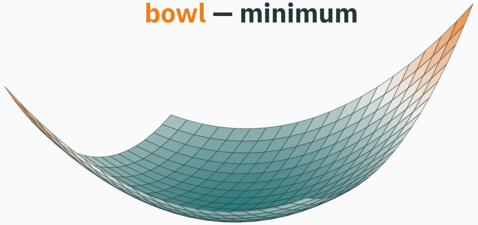
$f'' = 2 > 0$ – smile $f'' = -2 < 0$ – frown $f'' = 6x = 0$ – inconclusive

All three have $f'(0) = 0$. The **second** derivative tells them apart — except when it's 0, where the test goes silent (x^3: neither a min nor a max).

Hessian eigenvalues classify the local shape

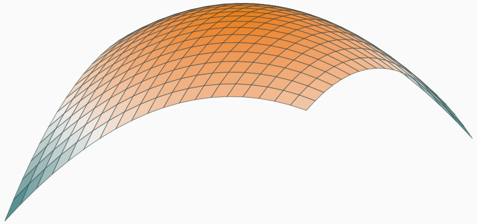
V VISUAL

bowl — minimum



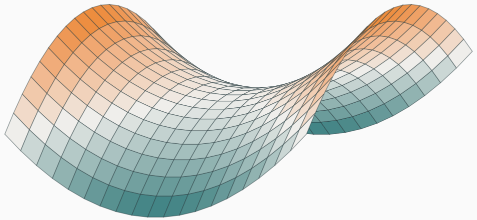
$\lambda = 3, 1$ – all > 0

dome — maximum



$\lambda = -1, -3$ – all < 0

saddle — neither



$\lambda = 2, -2$ – mixed signs

Curvature became a matrix in L9; its **eigenvalue signs** are the shape. (Eigenvalues above: computed live by the in-slide solver.)

> **At a stationary point: H's eigenvalue signs name the flat spot — all $+$ min, all $-$ max, mixed $=$ saddle.**

The Hessian-eigenvalue test

eigenvalues of H at $\nabla f = 0$	**classification**	**descent implication**
all $\lambda_i > 0$ (PD)	local **minimum**	park — locally, done
all $\lambda_i < 0$ (ND)	local **maximum**	any direction escapes
mixed signs (indefinite)	**saddle**	escape along a $\lambda < 0$ eigenvector
some $\lambda_i = 0$	**test is silent**	look beyond second order (x^3, troughs)

Hand tests for 2×2 still apply (L9): $\det H < 0 \to$ saddle; $\det H > 0$ — read the trace.

This test is local. Positive Hessian eigenvalues certify a local minimum, not a global one.

Worked example: two stationary points

$$f(x,y) = x^3 - 3x + y^2 \qquad \nabla f = \begin{pmatrix} 3x^2 - 3 \\ 2y \end{pmatrix} = \mathbf{0} \quad \Rightarrow \quad (x,y) = (\pm 1, 0)$$

Hessian: $H = \begin{pmatrix} 6x & 0 \\ 0 & 2 \end{pmatrix}$ — depends on where you stand (curvature is local):

- At $(1,0)$: $H = \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$, $\lambda = 6, 2$ — both positive $\rightarrow$ **local minimum**
- At $(-1,0)$: $H = \begin{pmatrix} -6 & 0 \\ 0 & 2 \end{pmatrix}$, $\lambda = -6, 2$ — mixed $\rightarrow$ **saddle**

And since $f(x,0) = x^3 - 3x \to -\infty$ as $x \to -\infty$: that “local minimum” is **not** global — nothing is. Flat-spot analysis alone can never tell you this.

Classify both points in NumPy

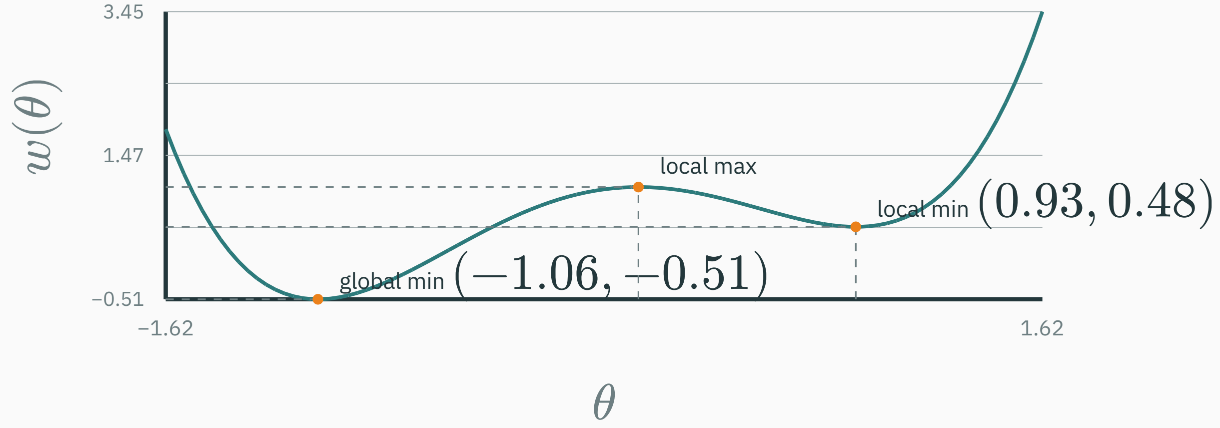
```
import numpy as np

def hess(x, y):                              # for f = x**3 - 3*x + y**2
    return np.array([[6*x, 0.0], [0.0, 2.0]])

for p in [(1, 0), (-1, 0)]:
    lam = np.linalg.eigvalsh(hess(*p))          # symmetric -> real, sorted
    kind = "min" if lam[0] > 0 else ("max" if lam[-1] < 0 else "saddle")
    print(p, lam, kind)
# (1, 0)  [2. 6.]  min
# (-1, 0) [-6. 2.] saddle
```

`eigvalsh` is the same symmetric-eigenvalue calculation used in L9; here the signs determine the local classification.

A local minimum need not be global

V VISUAL

Consider the non-convex function $w(x) = (x^2 - 1)^2 + x/2$. Every marked value below is computed in-slide using Newton's method on $w' = 0$:



- **Local min**: no better point in some ball around it. **Global min**: none anywhere.
- Descend from $\theta = 1.5$ and you park at 0.93 — flat, curving up — yet 1.00 above the global minimum.

The values of f, ∇f, and H at 0.93 do not reveal the lower minimum at -1.06.

Checkpoint: what do local derivatives certify?

Q QUESTION

An optimizer stops at $\hat{\theta}$ on a smooth, unknown $f : \mathbb{R}^2 \to \mathbb{R}$, with $\nabla f(\hat{\theta}) = 0$ and $H(\hat{\theta})$ having eigenvalues 2 and 5. What is certain?

A. $\hat{\theta}$ is a local minimum — and that is all we can certify

B. $\hat{\theta}$ is the global minimum

C. $\hat{\theta}$ is a saddle point

D. f is convex

Answer: local derivatives certify a local minimum

A ANSWER

Correct: A — a local minimum — and nothing more

Why: Both eigenvalues positive $\to$ PD $\to$ a genuine local min (L9). But **global** (B) is a claim about all of $\mathbb{R}^2$, and local probes at one point can't reach that far — the two-valley example $w(x)$ just demonstrated it. (D) reverses today's logic: curvature at **one** point never certifies convexity — x^3 has $f''(1) = 6 > 0$ yet isn't convex. Turning "local min" into "global min" needs a **property of f**, not a better probe. Enter convexity.

Convex sets: segments that never leave

The fix must be a property of shapes

What we want is a **certificate**:

$$\underbrace{\text{local conditions}}_{\nabla f=\mathbf{0}\ ,\quad H\succeq 0} + \underbrace{\text{a property of } f \text{ as a whole}}_{?} \quad\Rightarrow\quad \text{global claim}$$

That property is **convexity**, and it comes in two pieces, buildable in one lecture:

- **convex sets** — regions where straight segments never leave (this section)
- **convex functions** — surfaces whose chords never dip below the graph (next)

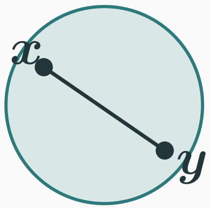
> Sets first because functions need somewhere to live: the chord test only makes sense if the segment between two inputs stays inside the domain.

The segment test

A set $C \subseteq \mathbb{R}^d$ is **convex** if for all $\boldsymbol{x}, \boldsymbol{y} \in C$ and every $t \in [0, 1]$:

$$t\boldsymbol{x} + (1-t)\boldsymbol{y} \in C$$

- $t\boldsymbol{x} + (1-t)\boldsymbol{y}$ is a point **on the segment**: $t = 0$ gives $\boldsymbol{y}$, $t = 1$ gives $\boldsymbol{x}$
- Convex = “walk between any two members and you never step outside”



A set is convex exactly when it contains every segment between two of its members.

Numbers: walking the segment

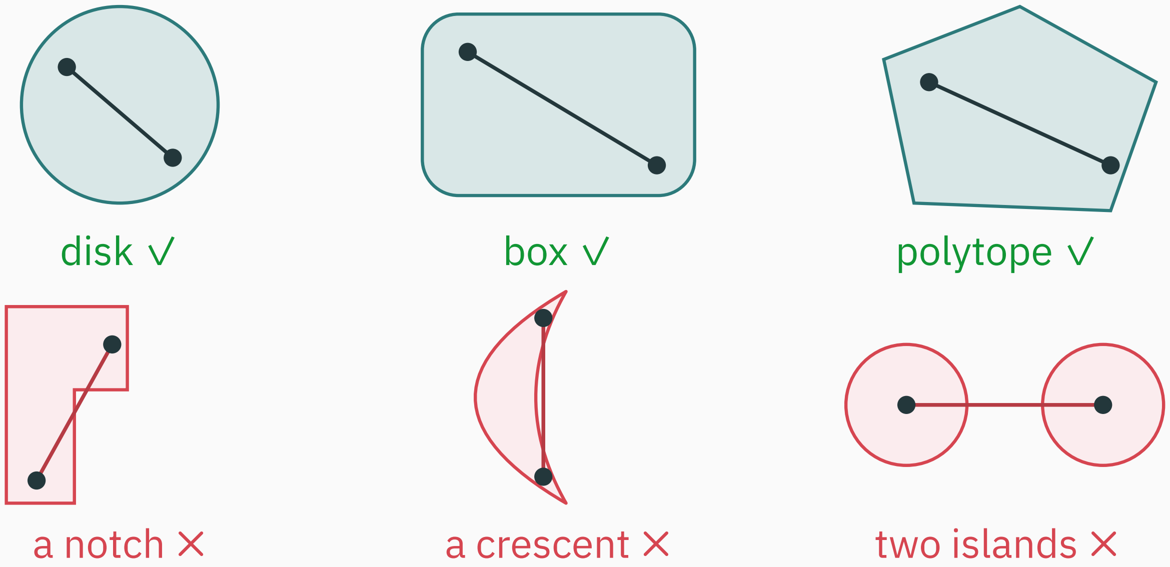
Take $\boldsymbol{x} = (4, 1)$ and $\boldsymbol{y} = (0, 3)$. The mix $t\boldsymbol{x} + (1 - t)\boldsymbol{y}$, by hand:

t	0	$1/4$	$1/2$	$3/4$	1
mix	$(0, 3)$	$(1, 2.5)$	$(2, 2)$	$(3, 1.5)$	$(4, 1)$

- $t = 0$ sits at $\boldsymbol{y}$, $t = 1$ at $\boldsymbol{x}$; in between, a straight walk — each coordinate is mixed independently
- The same expression with weights $(t, 1 - t)$ will reappear within minutes as the **chord** of a function — same algebra, one dimension up

Vocabulary you'll meet in Boyd: a **convex combination** is a mix with nonnegative weights summing to 1 — segments are the 2-point case.

Examples of convex and non-convex sets



One escaped segment settles it: convexity is a **for-all** claim; refuting it needs only one counterexample.

The convex sets ML lives on

set	where you meet it	convex?
$\mathbb{R}^d$ — all of parameter space	unconstrained training	✓
box $[0,1]^d$	valid pixel intensities	✓
half-space $\{\boldsymbol{x} : \boldsymbol{a}^\top \boldsymbol{x} \le b\}$	one side of a flat wall	✓
norm ball $\{\theta : \|\theta\| \le r\}$ (L3)	weight constraints, regularization	✓
simplex $\left\{\boldsymbol{p} : p_i \ge 0, \sum_i p_i = 1\right\}$	probability vectors (softmax)	✓ (L19)
sphere $\{\boldsymbol{x} : \|\boldsymbol{x}\| = 1\}$ — **skin** only	normalized embeddings	✗

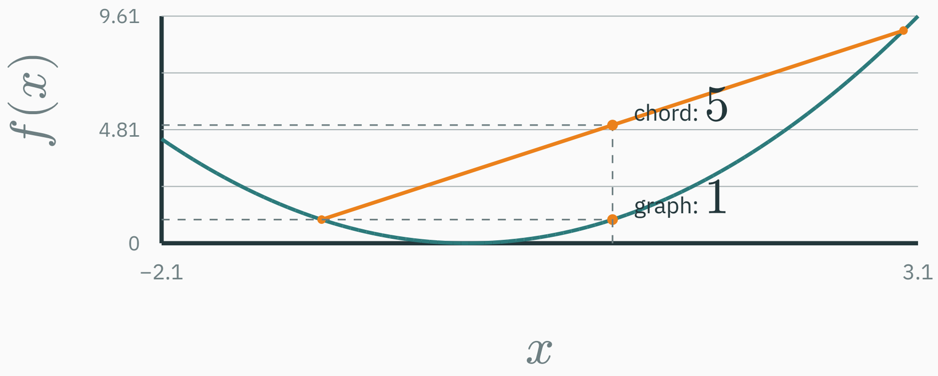
The sphere fails the test: mix two antipodes at $t = 1/2$ and you land at $\mathbf{0}$ — the center, off the skin. **Ball convex, shell not.**

Intersections preserve convexity (every fence respected) — L21's LP feasible regions are stacks of half-spaces. Unions, as the two islands showed, are not.

Convex functions: the chord test

Lift the segment to the graph

A function is **convex** when every chord — a segment joining two points **on the graph** — sits on or above it:



$$f\left(\underbrace{tx_1 + (1-t)x_2}_{\text{mix the inputs}}\right) \le \underbrace{tf(x_1) + (1-t)f(x_2)}_{\text{mix the outputs = chord height}} \qquad \forall x_1, x_2, t \in [0,1]$$

Convex: the graph hangs below its chords — on a convex domain.

Reading the inequality with numbers

$f(x) = x^2$, endpoints $x_1 = -1$, $x_2 = 3$. Mixed input: $tx_1 + (1-t)x_2 = 3 - 4t$.

t	0	1/4	1/2	3/4	1
mixed input $3-4t$	3	2	1	0	-1
graph $f(\text{mix}) = (3-4t)^2$	9	4	1	0	1
chord $tf(-1) + (1-t)f(3) = 9 - 8t$	9	7	5	3	1

Graph $\leq$ chord in every column — with equality exactly at the endpoints. That is the definition, felt in integers.

Mnemonic: convex holds water (∪-shaped); concave is a cave (∩-shaped).

Two checks that need no calculus

$f(x) = x^2$, by pure algebra (the chord inequality, expanded):

$$\mathrm{LHS} - \mathrm{RHS} = (tx_1 + (1-t)x_2)^2 - tx_1^2 - (1-t)x_2^2 = -t(1-t)(x_1 - x_2)^2 \leq 0 \quad \checkmark$$

($t(1-t) \geq 0$ on $[0,1]$; a square is ≥ 0. Multiply out once at home — three lines.)

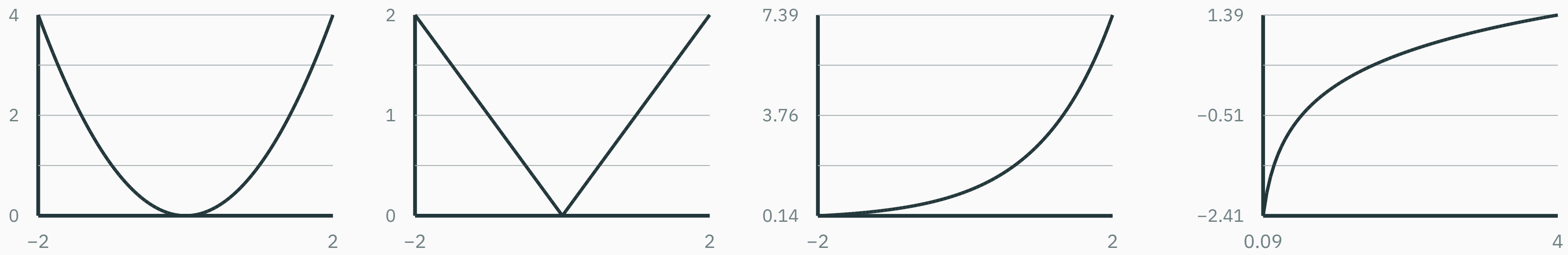
$f(x) = |x|$, by the triangle inequality (L3):

$$|tx_1 + (1-t)x_2| \leq |tx_1| + |(1-t)x_2| = t|x_1| + (1-t)|x_2| \quad \checkmark$$

The chord definition never mentions derivatives — it happily certifies the kinked $|x|$, where $f''(0)$ does not even exist.

The 1-D gallery, with verdicts

V VISUAL



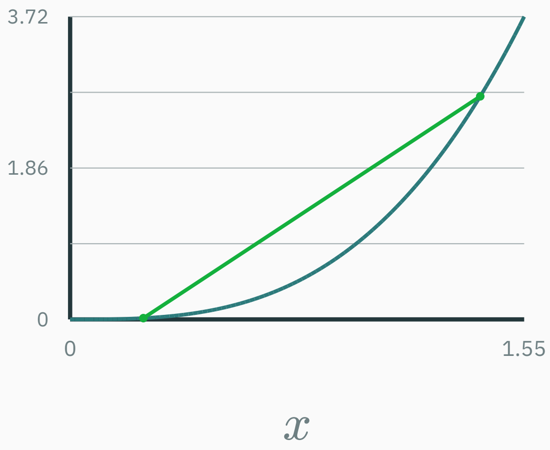
x^2 — convex on $\mathbb{R}$ | $|x|$ — convex on $\mathbb{R}$ | e^x — convex on $\mathbb{R}$ | $\ln x$ — **concave** on $(0, \infty)$

- e^x: convex even though it never turns around — convexity is about **bending up**, not about having a minimum
- $\ln x$: bends **down** everywhere — concave; and $-\ln x$ is convex (the shape of every NLL's building block, L14)

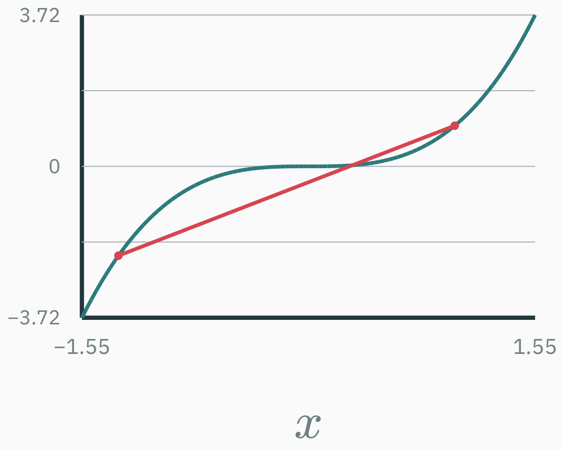
x^3 · the function that teaches domains

V VISUAL

on $[0, \infty)$: convex ✓



on $\mathbb{R}$: the chord **crosses** ×



- On $[0, \infty)$ every chord clears the graph; on $(-\infty, 0]$ it's concave; on all of $\mathbb{R}$, neither — the right chord dips below **and** rises above
- "Is f convex?" is ill-posed until you say **on which (convex) set** — same lesson the optimizers will learn about where they are allowed to search

Concave functions reverse the inequality

f is **concave** $\Leftrightarrow -f$ is convex $\Leftrightarrow$ chords hang **below** the graph.

Module 3 already used concavity:

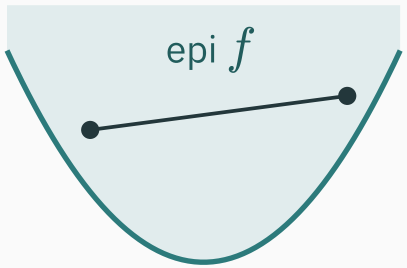
$$\underbrace{\max_\theta \ell(\theta)}_{\text{maximize log-likelihood (L14)}} \qquad = \qquad \underbrace{\min_\theta \mathrm{NLL}(\theta)}_{\text{minimize a }\textbf{convex}\text{ loss — often}}$$

- The coin's $\ell(\theta) = 7\ln\theta + 3\ln(1-\theta)$: sum of concaves — concave; its NLL is convex
- Gaussian $\ell(\mu)$: a downward parabola in μ (L14 showed it) — concave; MSE convex

Maximizing a concave likelihood is equivalent to minimizing its convex negative log-likelihood.

The epigraph ties sets and functions together

The two halves of today are the same object. The **epigraph** of f is everything on or above the graph:



$$\operatorname{epi} f = \{(\boldsymbol{x}, v) : v \geq f(\boldsymbol{x})\}$$

- f is a convex **function** $\Leftrightarrow$ $\operatorname{epi} f$ is a convex **set**
- A chord's endpoints are in the epigraph, so the whole chord must be — and "chord stays in epi f" **is** "chord above graph"

Boyd covers this bridge in Ch 3.1.7 — one segment test runs the entire theory. Nothing new to memorize.

Recognizing convexity without chords

Chords are the definition — calculus is the shortcut

Checking chords for every pair is direct but slow. For a twice-differentiable function on an open interval, curvature settles it:

$$f \text{ convex} \quad \Leftrightarrow \quad f''(x) \geq 0 \text{ everywhere on the interval}$$

The gallery, re-certified in one line each:

f	f''	**classification**
x^2	$2 > 0$	convex on $\mathbb{R}$ (and **strictly** — no flat stretches)
e^x	$e^x > 0$	convex on $\mathbb{R}$
$\ln x$	$-1/x^2 < 0$	concave on $(0, \infty)$
x^3	$6x$ — changes sign	convex on $[0, \infty)$ only
$\|x\|$	undefined at 0	test silent — the chord test already certified it convex ✓

In d dimensions: the Hessian test

$$f \text{ convex on } C \quad \Leftrightarrow \quad H(\theta) \succeq 0 \quad \text{for every } \theta \in C$$

Worked: $f(\theta_1, \theta_2) = \theta_1^2 + \theta_2^2$.

$$H = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \quad \text{everywhere} \quad \Rightarrow \quad \lambda = 2, 2 > 0 \quad \Rightarrow \quad \text{convex (strictly)} \checkmark$$

And the whole quadratic family at once: $q(\theta) = \frac{1}{2}\theta^\top A\theta$ has $H = A$ everywhere (L9′s ⭐) — so **q convex $\Leftrightarrow$ $A \succeq 0$**: bowls and troughs yes, saddles and domes no. The shape gallery was a convexity gallery all along.

Assumptions: C is open and convex and f is twice continuously differentiable on C. One PD Hessian at one point proves nothing (x^3 at $x = 1$, again).

The composition rules: convexity is stackable

Convex pieces snap together — for f, g convex:

rule	**why (one breath)**
αf for $\alpha \geq 0$	scaling both sides of the chord inequality preserves it
$f + g$	add the two chord inequalities
$f(A\boldsymbol{x} + \boldsymbol{b})$ — **affine into convex**	an affine map sends segments to segments, then f's chords take over
$\max(f, g)$	the upper envelope: each chord clears whichever graph is on top

Products and general compositions do not preserve convexity: $(x-1)^2 \cdot (x+1)^2 = (x^2-1)^2$ — two convex parabolas multiply into a non-convex double well.

Assemble the objectives you'll actually meet

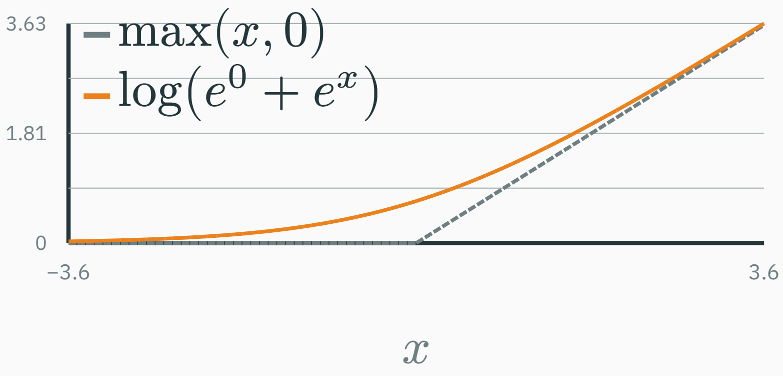
objective	build	convex?
$\|X\theta - \boldsymbol{y}\|^2$ — least squares	affine into $\|\cdot\|^2$	✓ (⭐ next)
$\|X\theta - \boldsymbol{y}\|^2 + \lambda\|\theta\|^2$ — ridge	sum of two convex terms	✓
$\|X\theta - \boldsymbol{y}\|^2 + \lambda\sum_i\|\theta_i\|$ — lasso	sum; $\|\cdot\|$ of affine	✓
$\max(0, 1 - z(\theta))$, z affine — hinge	max(affine, constant)	✓ (SVMs)
deep net loss	compositions of compositions	✕ in general

A **convex** penalty with a nonnegative weight preserves convexity; a non-convex regularizer can destroy it.

Real objectives are rarely chord-checked: assemble them from certified parts — that is how convexity scales.

Log-sum-exp is convex

L2 derived $\mathrm{LSE}(\boldsymbol{x}) = \log \sum_i e^{x_i}$ for numerical survival. It is also **convex**:



- A **smooth max**: hugs $\max(x, 0)$ without the kink — max's convexity survives the smoothing
- Why convex: $H = \mathrm{diag}(\boldsymbol{p}) - \boldsymbol{p}\boldsymbol{p}^\top$, $\boldsymbol{p} = \mathrm{softmax}(\boldsymbol{x})$, and $\boldsymbol{v}^\top H \boldsymbol{v} = \mathrm{Var}_{i\sim\boldsymbol{p}}[v_i] \geq 0$ — a variance (L12) cannot be negative

Logistic regression minimizes $\mathrm{LSE} - \mathrm{affine}$ — so it, too, is convex.

Checkpoint: certify or refute

Q QUESTION

Which one is not convex on all of $\mathbb{R}^2$, $\theta = (\theta_1, \theta_2)$?

A. $\|X\theta - \boldsymbol{y}\|^2 + 5\|\theta\|^2$

B. $\max\big(\theta_1 + \theta_2, e^{\theta_1}\big)$

C. $\log\big(e^{\theta_1} + e^{\theta_2}\big)$

D. $\theta_1^2 - \theta_2^2$

Answer: certify or refute

A ANSWER

Correct: D — $\theta_1^2 - \theta_2^2$ — a saddle, not a bowl

Why: Its Hessian is $\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$ everywhere — mixed eigenvalue signs, indefinite, the gallery's saddle: not convex (and not concave). The rest assemble from certified parts: (A) sum of convex ✓; (B) max of an affine and a convex ✓; (C) log-sum-exp ✓. Note how you refuted D and certified A–C **without one chord** — the toolkit is the point.

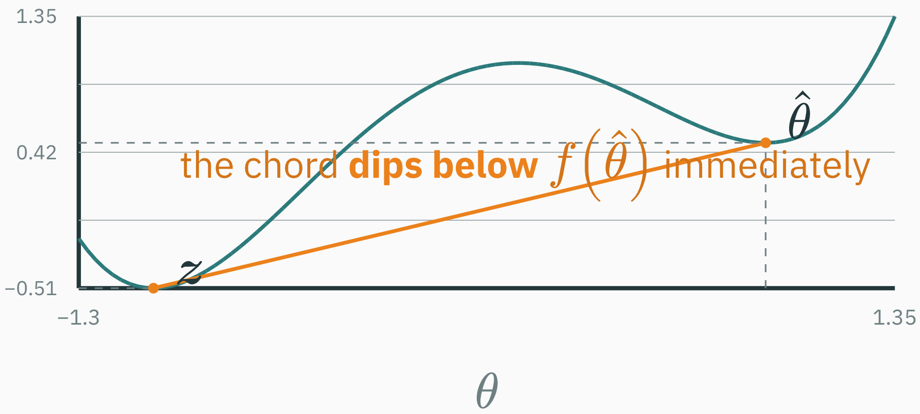
Local and global minima of convex functions

⭐ Every local minimum of a convex function is global

D DERIVATION

If f is convex, every local minimum is a global minimum.

Proof by picture first — suppose a convex f had a local min $\hat{\theta}$ **and** a strictly better point z:



The chord from $\left(\hat{\theta}, f\left(\hat{\theta}\right)\right)$ to $(z, f(z))$ slopes downward, and convexity keeps the graph on or below it — so points arbitrarily close to $\hat{\theta}$ are lower, contradicting local minimality.

⭐ The same picture, in four lines of algebra

D DERIVATION

Setup. f convex; $\hat{\theta}$ a local min; suppose some z has $f(z) < f\left(\hat{\theta}\right)$.

Walk toward z. For $t \in (0, 1]$, the mix $\theta_t = (1 - t)\hat{\theta} + tz$ lies on the segment. Convexity:

$$f(\theta_t) \leq (1 - t)f\left(\hat{\theta}\right) + tf(z)$$

Use the bad news. Since $f(z) < f\left(\hat{\theta}\right)$:

$$f(\theta_t) < (1 - t)f\left(\hat{\theta}\right) + tf\left(\hat{\theta}\right) = f\left(\hat{\theta}\right)$$

Contradiction. As $t \to 0$, θ_t enters **every** ball around $\hat{\theta}$ — yet each θ_t is strictly lower. No ball certifies $\hat{\theta}$ as a local min. ■

⭐ Least squares is convex

D DERIVATION

The objective L4 set up, L6 solved at scale, L14 crowned as Gaussian NLL:

$$f(\theta) = \|X\theta - \boldsymbol{y}\|^2 = (X\theta - \boldsymbol{y})^\top (X\theta - \boldsymbol{y}) = \theta^\top X^\top X\theta - 2\boldsymbol{y}^\top X\theta + \boldsymbol{y}^\top \boldsymbol{y}$$

Differentiate with L9′s two earned rules ($\nabla\, \theta^\top A\theta = 2A\theta$ for symmetric A; $\nabla\, \boldsymbol{b}^\top \theta = \boldsymbol{b}$):

$$\nabla f(\theta) = 2X^\top X\theta - 2X^\top \boldsymbol{y} \qquad H = \nabla(\nabla f) = 2X^\top X \quad \text{— constant: one bowl everywhere}$$

Convexity now hangs on one question: is $X^\top X$ **positive semi-definite for every possible data matrix** X?

⭐ The one-line certificate: $X^\top X \succeq 0$, always

D DERIVATION

Take any direction $v \in \mathbb{R}^d$ and test the quadratic form (L9′s definition of PSD):

$$v^\top (X^\top X)\, v = (Xv)^\top (Xv) = \|Xv\|^2 \geq 0 \quad \checkmark$$

A squared length cannot be negative, so $H = 2X^\top X \succeq 0$ for any data: least squares is convex, and every stationary point is a global minimizer.

On the tiny dataset $x = (0, 1, 2)$, $y = (1, 2, 4)$ (computed in-slide):

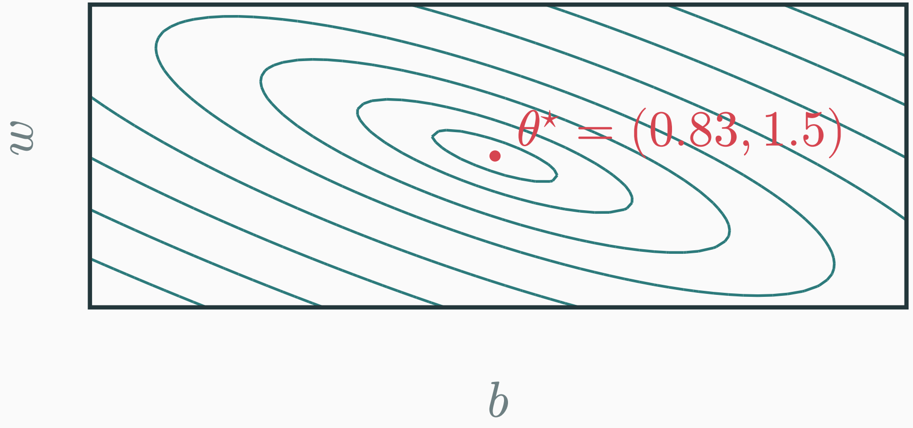
$$X^\top X = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \quad \lambda = 7.16, 0.84 > 0, \quad \nabla f = \mathbf{0} \Rightarrow \theta^\star = \begin{pmatrix} 0.83 \\ 1.50 \end{pmatrix}$$

Both eigenvalues are strictly positive, so this objective is strictly convex and $\theta^\star$ ($\hat{y} = 1.5x + 0.83$) is its unique minimizer.

The certified bowl, seen from above

V VISUAL

$f(b,w) = \sum_i (b + wx_i - y_i)^2$ over the whole parameter plane — one basin, one bottom:



- Elongated, tilted — but **one** valley: every downhill path drains to $\theta^\star$ (L17 walks it; the tilt and stretch will matter there)
- When columns of X are linearly dependent (L4′s rank!), an eigenvalue of $X^\top X$ hits 0: the bowl grows a **flat trough** of equally-good minimizers — still convex, still all-global, just no longer unique

Compare the two fitting problems

Why does the linear model "just work"?

- Least squares, ridge, lasso, and logistic regression have convex objectives under their usual formulations: local minima are global. An algorithm still needs suitable step sizes or other convergence conditions.
- A deep network is **compositions of compositions** — the one operation the rule table refuses to bless. Its landscape genuinely holds valleys, ridges, saddles, plateaus
- Deep-network training therefore lacks the same global certificate and is more sensitive to initialization and optimization choices.

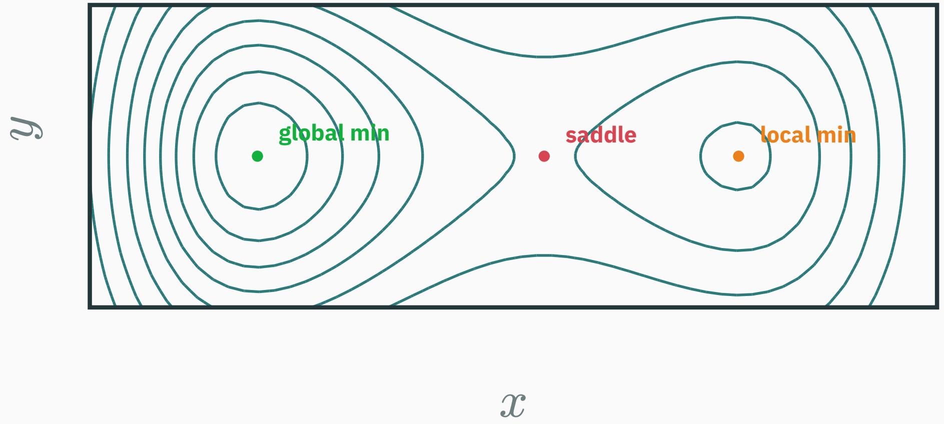
Convexity gives a global guarantee from local optimality. Non-convex objectives do not provide that guarantee.

Non-convex objectives

Several features of a non-convex objective

V VISUAL

The double well's 2-D sibling, $R(x, y) = (x^2 - 1)^2 + x/2 + 1.2y^2$ — all three flavors of flat spot in one place (marks placed by the in-slide Newton solver):



- The 1-D local **max** at 0.13 became a 2-D **saddle** (down along x, up along y) — extra dimensions turn maxima into saddles
- Two basins, one ridge between: where you land is decided by where you start

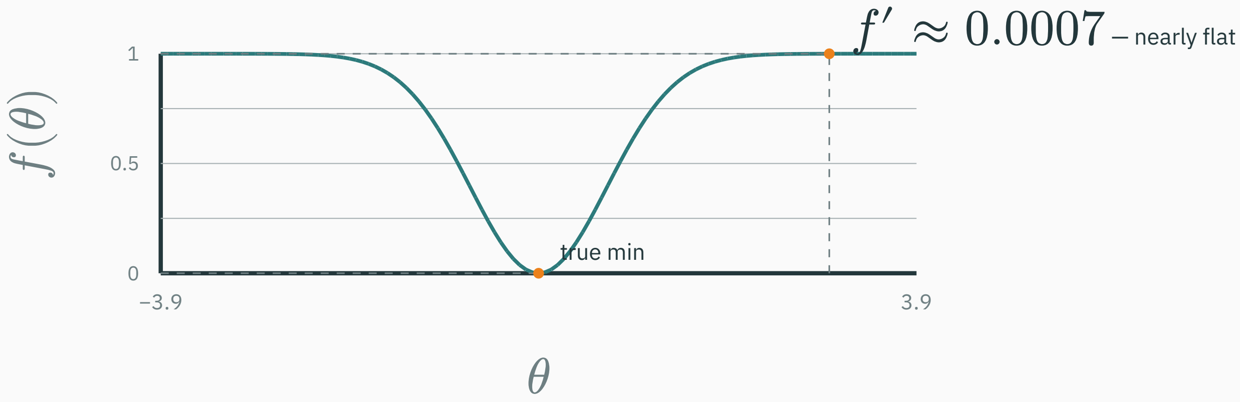
A counting heuristic for curvature signs

Toy model only: suppose the d Hessian eigenvalue signs were independent fair coin flips. Then the chance of all-positive curvature would be 2^{-d}:

d	$P(\text{all } \lambda > 0) \approx 2^{-d}$	**meaning**
2	0.25	minima common in toy problems
20	$\approx 10^{-6}$	minima already rare
10^6	$\approx 10^{-301030}$	all-positive is negligible under this toy model

- A saddle has a $\lambda < 0$ eigen-direction that provides a local descent direction; saddles often slow first-order methods without being local minima
- Real Hessian eigenvalues are neither independent nor fair coin flips, so this table is not a probability model for neural-network stationary points. It only illustrates why mixed signs have many combinatorial patterns in high dimension.

Plateaus: small gradients away from an optimum



- On a **plateau**, $\nabla f \approx \mathbf{0}$ with no minimum nearby: steps shrink to a crawl — the failure mode behind “training is stuck” and saturated activations (DL course)
- Neural-network training still finds useful parameters; explaining when and why needs assumptions beyond this lecture.

Convex objectives support global certificates; for non-convex ones, local tests remain local.

Lecture 16 — summary

- **One problem shape**: $\min_\theta f(\theta)$ — probed through f, ∇f, sometimes H.
- **Stationary points** $\nabla f = \mathbf{0}$: H's eigenvalue signs classify them (L9).
- **Convex set**: segments never leave. **Convex function**: chords never dip below.
- **Certify fast**: $f'' \geq 0$ / $H \succeq 0$ everywhere, or assemble — scale, sum, affine, max.
- ⭐ **Convex ⇒ local = global** — the four-line chord contradiction.
- ⭐ **Least squares is convex**: $H = 2X^\top X$ and $\boldsymbol{v}^\top X^\top X \boldsymbol{v} = \|X\boldsymbol{v}\|^2 \geq 0$ — for any data.
- **Without convexity**: saddles and plateaus; local need not be global.

Read before L17 — MML §7.3–7.3.1 (convexity, about 4 pages) · Boyd & Vandenberghe, *Convex Optimization* (free PDF): **skim** §2.1–2.3 and §3.1.1–3.1.5 + 3.2 — figures and boxed statements first. Distill's momentum article waits for L17.

Practice problems

Try on paper; anything numeric can be checked with five lines of NumPy.

1. Chord-check $f(x) = |2x - 1|$: prove convexity from the definition (triangle inequality + the affine trick).
2. Find **all** stationary points of $f(x, y) = x^4 + y^4 - 4xy$ and classify each with the Hessian. Which are global? (Careful: there's a tie.)
3. Is $g(x) = xe^x$ convex on $\mathbb{R}$? On which interval is it convex? ($g'' = (x + 2)e^x$.)
4. Ridge: show $H = 2X^\top X + 2\lambda I$ and explain why $\lambda > 0$ makes H positive **definite** even when $X^\top X$ has a zero eigenvalue. What does that do to the trough of equally-good minimizers?
5. Prove or refute with a picture: (a) the intersection of two convex sets is convex; (b) the union is.
6. For $d = 2$, evaluate log-sum-exp's Hessian $\mathrm{diag}(\boldsymbol{p}) - \boldsymbol{p}\boldsymbol{p}^\top$ at $\boldsymbol{x} = (0, 0)$, find the eigenvector with $\lambda = 0$, and connect it to L2's subtract-the-max trick.

⭐⭐⭐ Jensen's inequality: chords, industrialized

★ OPTIONAL

Run the chord inequality with n points instead of 2 (induction), weights $t_i \geq 0$, $\sum t_i = 1$:

$$f\left(\sum_i t_i x_i\right) \leq \sum_i t_i f(x_i) \qquad \text{and with } t_i = \text{probabilities:} \quad f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

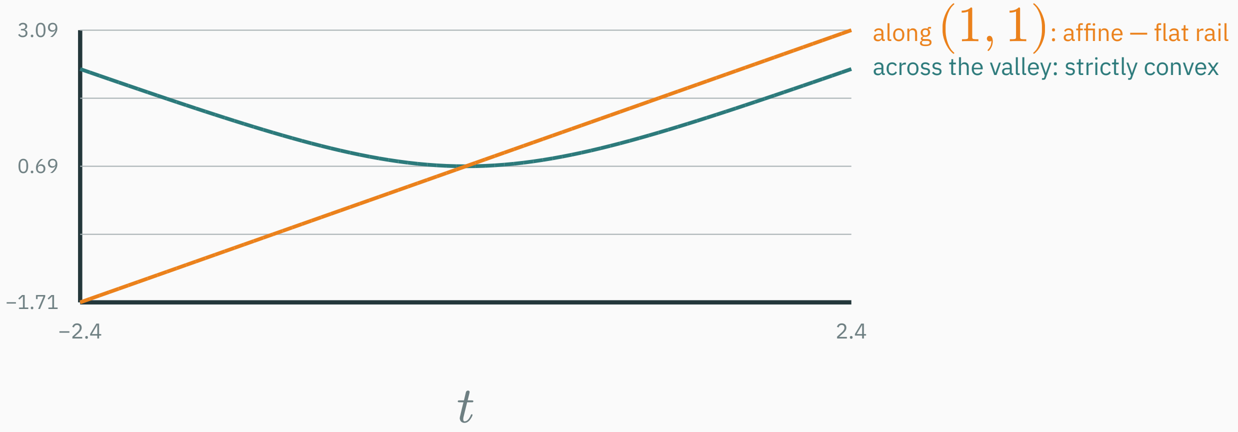
- For **concave** f, flip: $\mathbb{E}[\log X] \leq \log \mathbb{E}[X]$ — the single most-used inequality in probabilistic ML
- Module 5 runs on it: entropy bounds, $\mathrm{KL} \geq 0$, the ELBO — all one chord, industrialized (L22–L24)

> Weights on the simplex — the convex set from today's gallery. The definitions keep meeting.

⭐⭐⭐ LSE's flat valley is L2′s shift-invariance

★ OPTIONAL

Log-sum-exp is convex but not **strictly**: its bowl has one perfectly flat rail.



- Along $\mathbf{1} = (1, \ldots, 1)$: $\mathrm{LSE}(\boldsymbol{x} + c\mathbf{1}) = \mathrm{LSE}(\boldsymbol{x}) + c$ — affine, zero curvature; indeed $H\mathbf{1} = (\mathrm{diag}(\boldsymbol{p}) - \boldsymbol{p}\boldsymbol{p}^\top)\mathbf{1} = \boldsymbol{p} - \boldsymbol{p} = \mathbf{0}$
- That flat direction **is** L2′s softmax shift-invariance — subtract-the-max just slides along the rail where nothing changes. Geometry and numerics, one fact.

For a convex objective, every local minimum is global.

Next: **Gradient Descent** — now we actually walk downhill: trust the linear model for one small step, and repeat.